

Real-Time Optical and Acoustic Characterization of Phase-Shift Droplets for Drug-Delivery Applications

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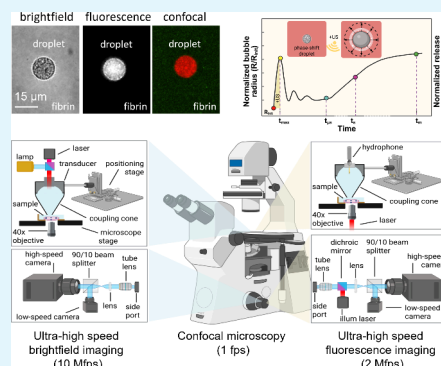
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ABSTRACT: Phase-shift droplets undergoing acoustic droplet vaporization (ADV) offer a promising approach for ultrasound-mediated drug delivery, enabling the spatiotemporally controlled release of therapeutic payloads. A comprehensive understanding of their behavior, through both optical and acoustic methods, is essential for optimizing the therapeutic efficacy. In this study, we investigated the effects of driving pressure, pulse duration, and bulk boiling point of perfluorocarbon droplets on ADV dynamics, payload release, and acoustic emissions. We employed ultra-high-speed brightfield [10 million frames per second (Mfps)], fluorescence (2 Mfps), and confocal microscopy (1 fps) to capture ADV and real-time payload release in fibrin-based hydrogels. During ADV, payload release velocities reached 2–4 m/s, slowing to 0.6–2.7 $\mu\text{m/s}$ post ultrasound. While the cycle number and pressure affected early bubble expansion and acoustic output, long-term bubble behavior and release kinetics were governed by the droplet's thermophysical properties. Ultra-high-speed imaging revealed a direct coupling between bubble dynamics and payload release continuing via diffusion after ultrasound. Notably, payload release rates post-ultrasound exceeded bubble growth rates. Additionally, acoustic emissions, recorded via passive cavitation detection, increased with both pressure and pulse number but decreased in droplets with higher bulk boiling points. These findings underscore the importance of integrating multimodal imaging methods to elucidate ADV mechanisms and design effective hydrogel-based drug-delivery systems.

KEYWORDS: phase-shift droplets, acoustic droplet vaporization, ultrahigh-speed imaging, drug delivery, ultrasound



1. INTRODUCTION

Controlled drug delivery has emerged as a powerful strategy to enhance therapeutic efficacy while minimizing systemic side effects. Conventional administration methods, such as oral ingestion or intravenous injection, often result in suboptimal drug concentrations at target sites and undesired off-target effects.¹ Drug-delivery systems enabling active control over release parameters—including dosage, temporal profile, spatial localization, and release kinetics—are gaining increasing interest for their potential in personalized and site-specific therapies.^{1,2} Despite progress, most local delivery systems still struggle to precisely regulate these parameters. Ultrasound-mediated drug delivery has shown promise in addressing these limitations. Already widely used in clinical imaging, ultrasound has proven its therapeutic utility in applications such as extracorporeal shock-wave lithotripsy³ and, more recently, histotripsy.^{4,5} Additionally, ultrasound is nonionizing, non-invasive, and capable of deep-tissue penetration, making it safe for repeated use and suitable for targets deep inside the body.^{6–8} Moreover, the ability of ultrasound to focus energy within a small focal volume—typically on the order of a few cubic millimeters at −6 dB—enables high spatial precision for targeted treatments.⁹

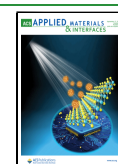
One extensively studied ultrasound-mediated drug-delivery mechanism involves contrast microbubbles.^{10,11} Therapeutic payloads can be incorporated in various ways, including encapsulated within the microbubble shell (e.g., dissolved in an oil layer between the gas core and shell or incorporated within the shell), attached to the shell surface via electrostatic interactions or bioconjugation strategies (e.g., avidin–biotin linkages), or coadministered with the microbubbles in solution.^{12–14} A more comprehensive review of microbubble-based drug-loading approaches can be found in ref 14. This method is particularly advantageous for delivering drugs that are difficult to accumulate at the target sites in sufficiently high concentrations without adverse effects. Microbubbles can facilitate localized delivery by leveraging ultrasound-induced cavitation, which can occur in either the stable or inertial regime depending on factors such as the excitation frequency, bubble size, and temperature.^{13,15} The mechanical index (MI)

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is commonly used as a practical indicator of cavitation: $MI < 0.2$ corresponds to linear scattering, $0.2 < MI < 0.7$ is associated with stable cavitation (nonlinear oscillations), and $MI > 0.7$ is generally indicative of inertial cavitation.^{16–18} However, microbubbles suffer from inherent limitations, including a small shell volume for drug loading, size constraints (typically 1–10 μm) that confine them to intravascular spaces, and a short circulation half-life on the order of minutes.^{19–21}

To overcome these limitations, phase-shift droplets have been developed as next-generation ultrasound-responsive agents for theranostic applications.²² These droplets consist of a liquid perfluorocarbon (PFC) core that vaporizes into gas when exposed to ultrasound above a certain pressure threshold in a process known as acoustic droplet vaporization (ADV).^{22,23} The ADV threshold is influenced by both intrinsic factors (e.g., bulk boiling point of the PFC core) and extrinsic parameters (e.g., ultrasound frequency). For instance, sub-micron-sized octafluoropropane droplets exhibit an ADV threshold below 500 kPa at 4.7 MHz at 37 °C.²⁴ In contrast, micron-sized perfluorooctane droplets require ~ 3 MPa at 2.5 MHz,²⁵ while micron-sized perfluorohexane droplets vaporize at ~ 5.5 MPa at 8.6 MHz.²⁶ During ADV, generated bubbles rapidly expand, with the final volume exceeding the initial droplet size. The dynamics of this expansion are influenced by several factors, including the acoustic driving conditions, the bulk boiling point of the PFC, and the viscoelastic properties of the surrounding medium.²⁷ Phase-shift droplets offer greater drug-loading capacity into the liquid core, resulting in therapeutically relevant dosages. Encapsulated drugs are expelled from the droplet as its structure is disrupted during vaporization.^{28,29}

ADV of intravascularly administered droplets has shown potential in delivering chemotherapeutics like paclitaxel for targeted treatment of ovarian, breast, and pancreatic cancers³⁰ as well as scavenging oxygen to mitigate myocardial reperfusion injury.³¹ One particular area of interest is hydrogel-based implants embedded with drug-loaded phase-shift droplets, in the form of double emulsions.³² These hydrogels serve as drug depots, protecting the biological activity of therapeutic agents, and have been used to induce localized, biological responses such as angiogenesis through the controlled release of growth factors,^{33,34} osteogenesis through bubble-mediated matrix compaction,³⁵ modulation of cellular signaling in a cancer model,³⁶ and cell differentiation.³⁷ Phase-shift droplets can be uniformly dispersed through the hydrogel³² or micropatterned at high spatial resolution via 3D bioprinting.³⁸ Recently, a composite hydrogel approach demonstrated a triphasic release profile using ADV.²⁹ Despite these advances, most evaluations of ADV-mediated payload release have relied on postexposure quantification methods such as flow cytometry, fluorescence spectrophotometry, laser speckle microscopy, and confocal imaging, measuring release hours or days after ultrasound exposure. Real-time visualization of ultrasound–droplet interactions is essential for a more mechanistic understanding of the release kinetics.

Optical microscopy techniques that span multiple temporal resolutions—from low frame rate confocal fluorescence imaging to ultrahigh-speed imaging [up to 25 million frames per second (Mfps)]—are critical to fully characterizing droplet dynamics and the induced biological effects. Ultra-high-speed imaging has previously deepened our understanding of microbubble phenomena, including shell buckling, compression-only oscillations, and shape oscillations.^{39–41} Several

groups have used this approach to visualize ADV across different time scales, capturing initial nucleation, rapid expansion, oscillatory behavior, and subsequent bubble growth via passive diffusion.^{42–47} These studies provided an understanding of the physics underlying nucleation and ADV dynamics and have been essential to the ongoing development of ADV for specific applications, ultimately leading to a more predictable, controlled, and efficient translation to preclinical and clinical studies.

One limitation of high-speed imaging is its reliance on brightfield microscopy, which captures the diffraction shadow image of the gas core but does not visualize the encapsulated payload. Due to insufficient contrast between the encapsulated payload and the surrounding media, brightfield microscopy cannot be used to track payload release kinetics. High-speed fluorescence microscopy, capable of visualizing payload dynamics during ultrasound exposure, remains largely unexplored—particularly at ultrafast frame rates. Previous studies using high-speed fluorescence microscopy have been predominantly used for conventional microbubbles, capturing dynamic processes such as lipid shedding from microbubble shells during ultrasound exposure, a proposed mechanism of drug release.^{39,48,49} A threshold oscillation amplitude ($\Delta R/R_0$) of $\sim 35\%$ was shown to trigger lipid shedding from microbubbles with an average diameter of ~ 3 μm . At an excitation frequency of 1 MHz, in-plane shedding occurred below 255 kPa, while out-of-plane shedding required pressures exceeding 425 kPa.⁵⁰

In this study, we present a comprehensive investigation of both ADV bubble dynamics and the corresponding payload release, spanning microsecond to minute time scales. We employed ultra-high-speed fluorescence microscopy at 2 Mfps to directly visualize and quantify real-time dynamics of payload release during ADV. Using a fluorescent hydrophilic tracer as a model payload, we analyzed key parameters, including the maximum release distance and release velocity. Additionally, ADV bubble dynamics were recorded via brightfield microscopy at 10 Mfps to characterize maximum radial expansion and expansion velocity. We also monitored acoustic emissions generated during ADV using passive cavitation detection. By studying different PFC droplets with varying bulk boiling points (29–100 °C), we evaluated how physicochemical properties influence release behavior. Through the integration of high-speed optical imaging, fluorescence microscopy, and acoustic analysis, this work aims to investigate the relationship among ADV-induced bubble dynamics, acoustic emissions, and payload release from single PFC droplets embedded in hydrogels.

2. MATERIALS AND METHODS

2.1. Preparation of Payload-Carrying Phase-Shift Droplets Using a Microfluidic Platform. Monodispersed droplets were fabricated using a flow-focusing microfluidic chip (channel dimensions: 14 $\mu\text{m} \times 17$ μm , Dolomite, Royston, U.K.) following previously established protocols.^{29,38} Because perfluorocarbons (PFCs) are hydrophobic and lipophobic, therapeutic agents are typically encapsulated in a secondary aqueous phase rather than dissolved directly in the PFC core.³² This configuration forms a water-in-PFC-in-water emulsion structure ($W_1/\text{PFC}/W_2$).^{32,51} Three different PFC liquids were used in this study: perfluoropentane (PFP, CAS 355-42-0, Strem Chemicals, Newburyport, MA, USA), perfluorohexane (PFH, CAS 355-42-0, Strem Chemicals), or perfluorooctane (PFO, CAS 307-34-6, Sigma-Aldrich, St. Louis, MO, USA). PFP [bulk boiling point (bbp): 29 °C at 1 atm] and PFH (bbp: 56 °C at 1 atm) have been frequently used for ultrasound and photoacoustic

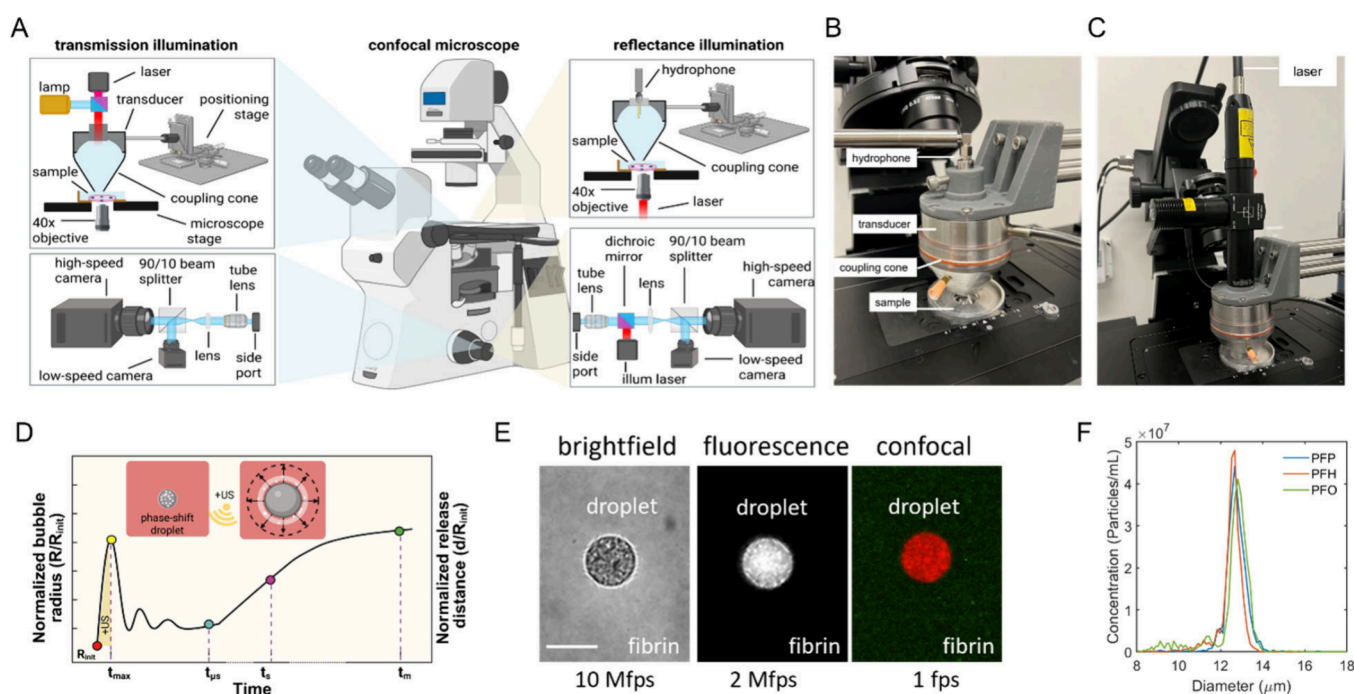


Figure 1. Multimodal imaging system for characterizing ADV dynamics and payload release. (A) Schematics of the transmission illumination setup used for ultra-high-speed brightfield imaging and the reflectance illumination setup used for ultra-high-speed fluorescence imaging. Both configurations were integrated into an inverted confocal microscope, enabling simultaneous optical and acoustic data acquisition. (B) Acoustic emissions captured using a hydrophone placed in the central opening of a single-element transducer. The orientation of the transducer with respect to the sample is also shown. In this panel, reflectance illumination was used. (C) Orientation of laser illumination in a transmission configuration. (D) Schematic timeline showing key imaging time points: during ultrasound (US) exposure [$t_{\max} = t(R=R_{\max})$], end of ultrahigh-speed imaging [$t_{\mu s} \sim O(\mu s)$], end of low-speed imaging [$t_s \sim O(s)$], and end of confocal time-lapse imaging [$t_m \sim O(\text{min})$]. The normalized bubble radius and payload release distance were extracted from recorded images across these time points. (E) Representative images of a phase-shift droplet embedded in a 10 mg/mL fibrin hydrogel recorded via ultra-high-speed brightfield, fluorescence imaging, and confocal microscopy. Alexa Fluor 647-labeled dextran was used as a fluorescent payload (red), and Alexa Fluor 488-labeled fibrinogen was used for fibrin visualization (green). (F) Concentration and average size distributions of three different PFC droplets. Scale bar: 15 μm . The panels in part A were created in Biorender.com.

imaging applications,^{52,53} while PFO (bbp: 100 °C at 1 atm) is often explored for therapeutic applications such as tissue engineering.^{25,54} The droplet fabrication involved two steps. First, an initial W_1 /PFC emulsion (1:2 volumetric ratio) was generated via sonication in an ice–water bath. The W_1 phase contained Alexa Fluor 647-labeled dextran (AF647, 10 kDa, Life Technologies, Grand Island, NY, USA) dissolved in phosphate-buffered saline (PBS, Life Technologies) and stabilized with a fluorinated surfactant. A fluorosurfactant copolymer was derived from a 2:1 M ratio of Krytox 157 FSH (CAS 51798-33-5, DuPont, Wilmington, DE, USA) and poly(ethylene glycol)bis(amine) (MW = 1000 g/mol, CAS 24991-53-5, Alfa Aesar, Ward Hill, MA, USA) and used at a concentration of 22.38 $\mu\text{g/mL}$ in the W_1 phase. The secondary (W_2) phase consisted of Pluronic F68 (50 mg/mL, CAS 9003-11-6, Sigma-Aldrich) in PBS. The W_1 phase was pumped by a syringe pump (KDS-410, KD Scientific, Holliston, MA, USA) through an UltraLine UHPLC in-line filter (0.5 μm stainless steel, Restek, Bellefonte, PA, USA) and into the inner channel of a quartz microfluidic chip. Concurrently, the W_2 phase was pumped by a second syringe pump (78-0388; KD Scientific, Holliston, MA, USA) through an in-line filter and into the outer channels of the chip. The flow rates in the inner and outer channels were 0.5 and 2.5 $\mu\text{L/min}$, respectively, to yield double emulsions with a nominal diameter of 13 μm . For a subset of studies, PFP droplets with varying concentrations of the W_1 phase were prepared for high-speed fluorescence microscopy to investigate self-quenching of droplets in a condensed state. Droplet size distributions were characterized using a Coulter Counter (30 μm aperture; Beckman Coulter, Brea, CA, USA), yielding average diameters of $12.4 \pm 0.4 \mu\text{m}$ (PFP droplets), $12.5 \pm 0.4 \mu\text{m}$ (PFH droplets), and $12.6 \pm 0.3 \mu\text{m}$ (PFO droplets).

2.2. Preparation of Fibrin-Based Tissue-Mimicking Hydrogels. Fibrin-based hydrogels were prepared as tissue-mimicking

substrates to study the behavior of phase-shift droplets under ultrasound exposure. Hydrogels were formulated by mixing bovine fibrinogen (10 mg/mL, Sigma-Aldrich), dissolved in FluoroBrite Dulbecco's modified Eagle's medium (Life Technologies), with recombinant human thrombin (2 U/mL, Recothrom, Baxter, Deerfield, IL, USA) and phase-shift droplets. A low droplet volume fraction [0.01% (v/v)] in the final hydrogel (interdroplet spacing $>200 \mu\text{m}$ ⁵⁵) was used to ensure that individual droplets could be visualized within the imaging field, while minimizing interdroplet interactions. Hydrogel polymerization was carried out directly within a custom-designed imaging chamber mounted on a confocal microscopy stage (Figure 1). For visualization of the fibrin network in confocal fluorescence imaging, Alexa Fluor 488-labeled fibrinogen (Molecular Probes, Eugene, OR, USA) was incorporated at a concentration of 39 $\mu\text{g/mL}$.

2.3. Ultrahigh-Speed Fluorescence and Brightfield Microscopy. An integrated optical–acoustic imaging system was built around an inverted confocal microscope (AX, Nikon, Melville, NY, USA) equipped with a 40 $\times$ water-immersion objective (PLAN APO, N.A. = 1.15, Nikon) to enable high-speed brightfield and fluorescence imaging during ultrasound exposure (Figure 1A). Illuminations for high-speed fluorescence imaging were provided by a pulsed laser (400 W, 640 nm; Cavitator Ltd., Tampere, Finland) with a total pulse duration of 30 μs . Fluorescence imaging was conducted in reflectance mode using a high-speed camera (HPV-X2; Shimadzu, Kyoto, Japan) at 2 Mfps with a resolution of 400×250 pixels. Additionally, a secondary camera (Teledyne FLIR, Wilsonville, OR, USA) was integrated to capture images at a lower frame rate of 500 fps. A nonpolarizing cube beam splitter [BS028-90:10 (R:T), Thorlabs, Newton, NJ, USA] divided the light path, directing most (90/10%) illumination to the high-speed camera and a smaller portion to the

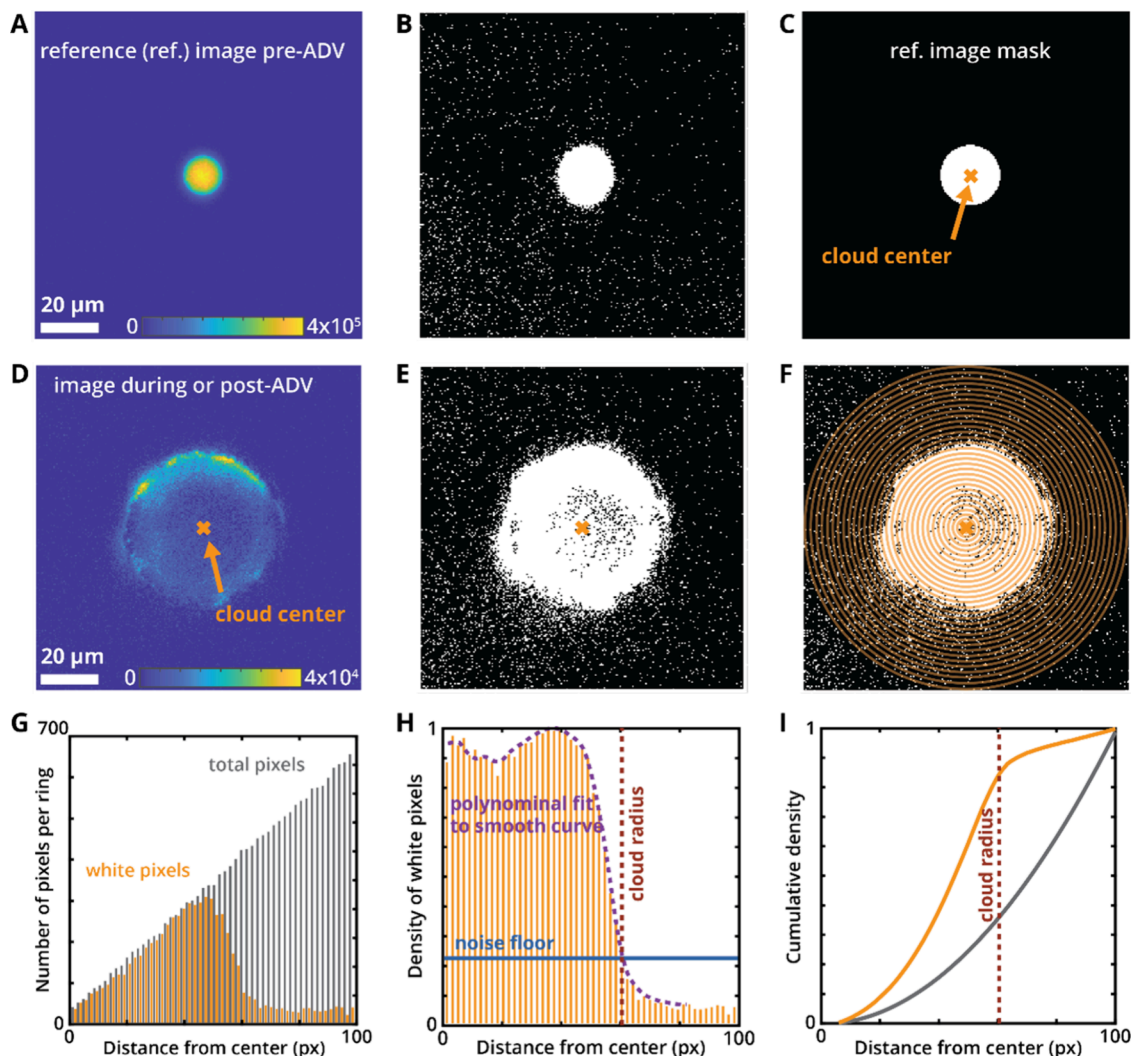


Figure 2. Image processing workflow for quantifying ADV-induced payload release dynamics. (A) Reference fluorescence image of a droplet recorded prior to ADV. (B) Binary representation of the reference frame generated by applying a threshold to isolate the droplet from background noise. (C) Reference image mask with the computed droplet center marked. (D) Fluorescence image acquired during or after ADV, showing the spatial spread of the released payload. (E) Binary representation of the post-ADV image generated using the noise threshold defined from the reference frame. (F) Concentric rings (1-pixel width) constructed around the cloud center to assess the radial intensity distribution. (G) Histogram showing the number of total pixels (gray) and threshold-exceeding white pixels (orange) in each radial ring. (H) Pixel density (white pixels per ring area) plotted as a function of the radial distance. A polynomial fit was applied to smooth the curve, and the cloud radius was determined as the point where the density falls to the noise floor (blue). (I) Cumulative density function of the payload signal as a function of the radial distance, with the cloud radius (dotted red line) indicating the spatial extent of release.

secondary camera, enabling synchronized, coaligned imaging across time scales. A Cy5 filter cube (Nikon) enabled the detection of fluorescence emitted from labeled droplets, facilitating payload release tracking. The orientation of the ultrasound transducer relative to the sample is shown in Figure 1B.

Brightfield imaging was performed at an ultrahigh frame rate of 10 Mfps (256 frames; 400×250 pixels), corresponding to a $25.6 \mu\text{s}$ imaging window. Transmission illumination mode was used for brightfield imaging (Figure 1A,C). Synchronization of the high-speed optical camera, laser illumination, and ultrasound transmission was accomplished using a function generator. To avoid repeated exposure of the same region, the imaging field was repositioned by at least $500 \mu\text{m}$, according to the lateral width of the beam at the focus, in the x and y directions after each ultrasound burst.

2.3.1. Image Processing and Analysis. Both ultrahigh-speed and low-speed fluorescence images were processed using custom scripts developed in MATLAB (MathWorks, Natick, MA, USA) to quantify the dynamics of payload release (Figure 2). In each case, the frame captured prior to ADV was designated as the reference image (Figure

2A). To delineate droplet boundaries in the reference frame (I_{ref}), a threshold of $\frac{1}{n} \sum_{i=1}^n I_{\text{ref}} + 2\sigma_{I_{\text{ref}}}$ was found to produce a consistent and repeatable binary mask (BW_{ref} ; Figure 2B), where $\sigma_{I_{\text{ref}}}$ is the reference image standard deviation. From this mask, the centroid of the droplet ($x_{\text{ref}}, y_{\text{ref}}$) was computed, and a background noise threshold (φ) was determined from the pixel intensity distribution outside the mask using the following expression:

$$\varphi = \frac{1}{n} \sum_{i=1}^n I_{\text{ref}}(\sim \text{BW}_{\text{ref}}) + 5\sigma_{\text{ref}}$$

where σ_{ref} represents the standard deviation of the background pixel intensities outside the droplet area (Figure 2C). This noise threshold served as a baseline for analyzing subsequent frames. For subsequent frames (Figure 2D), a new binary mask was generated based on the reference noise threshold (Figure 2E). The coordinates (x, y) of pixels in the binary mask were then used to compute their radial distance (r) from the reference droplet center using

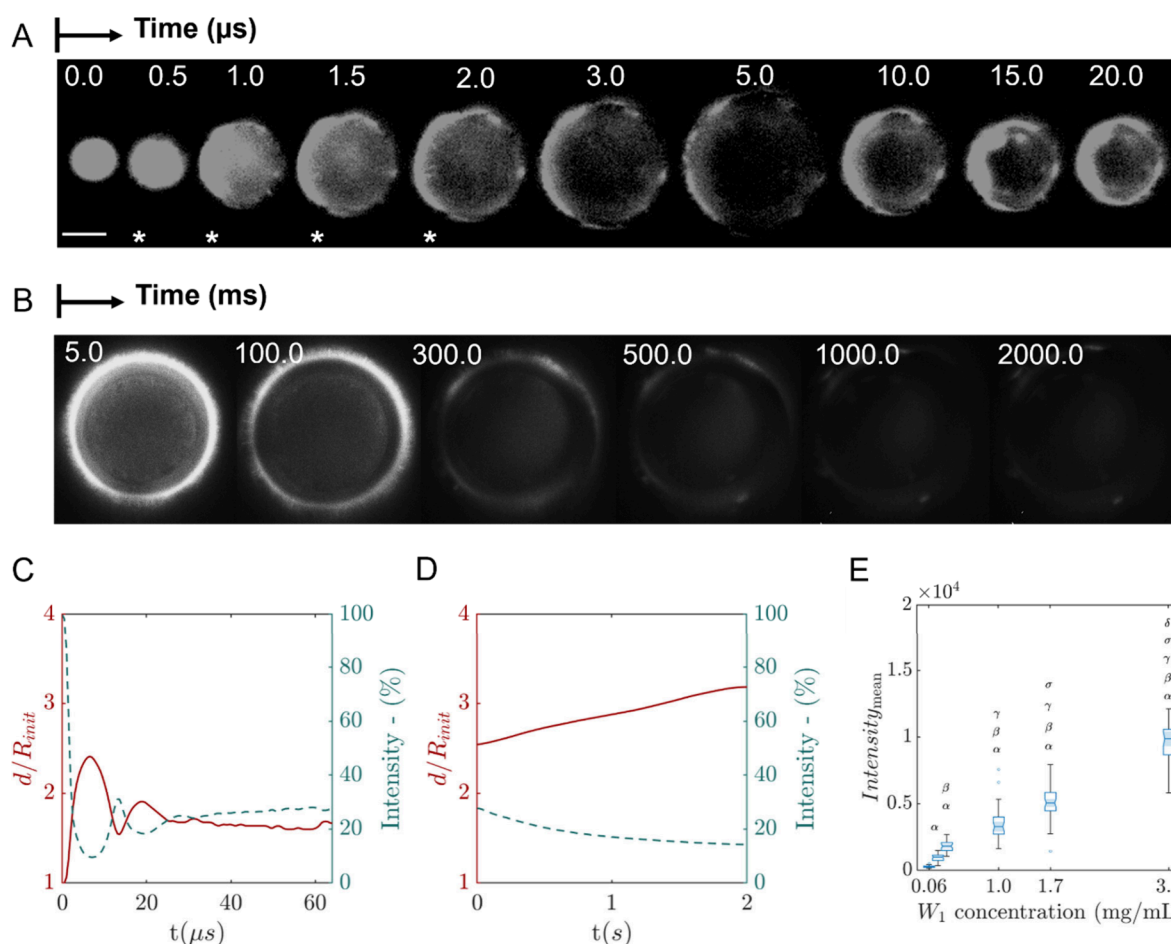


Figure 3. Temporal dynamics of payload release triggered by ADV in PFP phase-shift droplets. (A) Sequential fluorescence images acquired at 2 Mfps showing the temporal evolution of payload release before, during, and after ADV. A six-cycle burst of 2.5 MHz focused ultrasound at 6.5 MPa induced droplet vaporization, releasing the fluorescent payload. Ultrasound exposure is indicated by asterisks. (B) Fluorescence images captured over millisecond-to-second time scales, tracking post-ADV payload dispersal. Scale bar: 15 μm . (C and D) Quantitative analysis of normalized payload transport and intensity. d normalized to R_{init} is plotted on the left y axis, and the normalized fluorescence intensity is plotted on the right y axis. Data are presented for (C) microsecond and (D) second time scales. (E) Control studies were conducted on PFP droplets (in the absence of ultrasound) to quantify the mean signal intensity as a function of varying concentrations of the W_1 phase (i.e., Alexa Fluor 647-labeled dextran dissolved in PBS). Statistically significant differences ($p < 0.05$) are denoted as follows: α , vs 0.06; β , vs 0.18; γ , vs 0.3; σ , vs 1.0; δ , vs 1.7.

$$r = \sqrt{(x - x_{\text{ref}})^2 + (y - y_{\text{ref}})^2}$$

To quantify the spatial distribution of the released payload, a density function [$\rho(r)$] was defined by dividing the image into concentric rings (1-pixel width) centered on the droplet (Figure 2F). The total number of active pixels in each ring was normalized by the total number of pixels in that ring to generate the density profile (Figure 2G). The released payload radius (d)—defined as the radial extent of the fluorescence signal—was determined as the furthest distance from the center at which the normalized density function was greater than the noise threshold (τ) (Figure 2H), calculated as

$$\tau = \frac{1}{n} \sum_{i=1}^n I_{\text{ref}}[\rho(\rho < \varphi)] + 0.25\sigma_{\rho}$$

Plotting $\rho(r)$ versus radial distance (r) enabled detailed visualization of released payload expansion dynamics (Figure 2I). The release velocity [$V(d,t) = \Delta d/\Delta t$] was defined as the change in the front position of the payload over time step.

Ultrahigh-speed brightfield image sequences were processed using a separate edge detection-based algorithm in MATLAB to extract time-resolved bubble radii following ADV events, as has been done previously.⁵⁶

2.4. Time-Lapse Fluorescence Confocal Microscopy. To evaluate the postvaporization behavior of ADV-generated bubbles over extended time scales and characterize diffusion-driven payload release kinetics, time-lapse confocal fluorescence microscopy was employed. Imaging was performed at a frame rate of 1 fps over a total duration of 120 s. All experiments were conducted at a physiologically relevant temperature of 37 $^{\circ}\text{C}$ using a stage-mounted temperature control system (TCS-200; AmScope, Irvine, CA, USA). Confocal microscopy data were analyzed using the NIS-Elements AR 5.4 software (Nikon).

2.5. Ultrasound Setup and Parameters. To induce ADV, a calibrated, single-element-focused transducer (2.5 MHz, model H-147, diameter = 60 mm; radius of curvature = 50 mm, Sonic Concepts, Bothell, WA, USA) was mounted on a 3D micropositioning stage. The transducer featured a circular cutout in the center (20 mm diameter), enabling optical access for simultaneous high-speed imaging and acoustic measurements (Figure 1B,C).

The transducer was calibrated in a free field using a custom-built fiber-optic hydrophone (fiber diameter = 105 μm ; sensitivity = 16.6 mV/MPa) and a precision scanning system (AIMS III, Onda, Sunnyvale, CA, USA). The focal region had full-width at half-maximum dimensions of 4.7 ± 0.2 mm axially and 0.6 ± 0.2 mm laterally, as measured using a fiber-optic hydrophone in an automated scanning system (AIMS III, Onda, Sunnyvale, CA, USA). Hydrogel

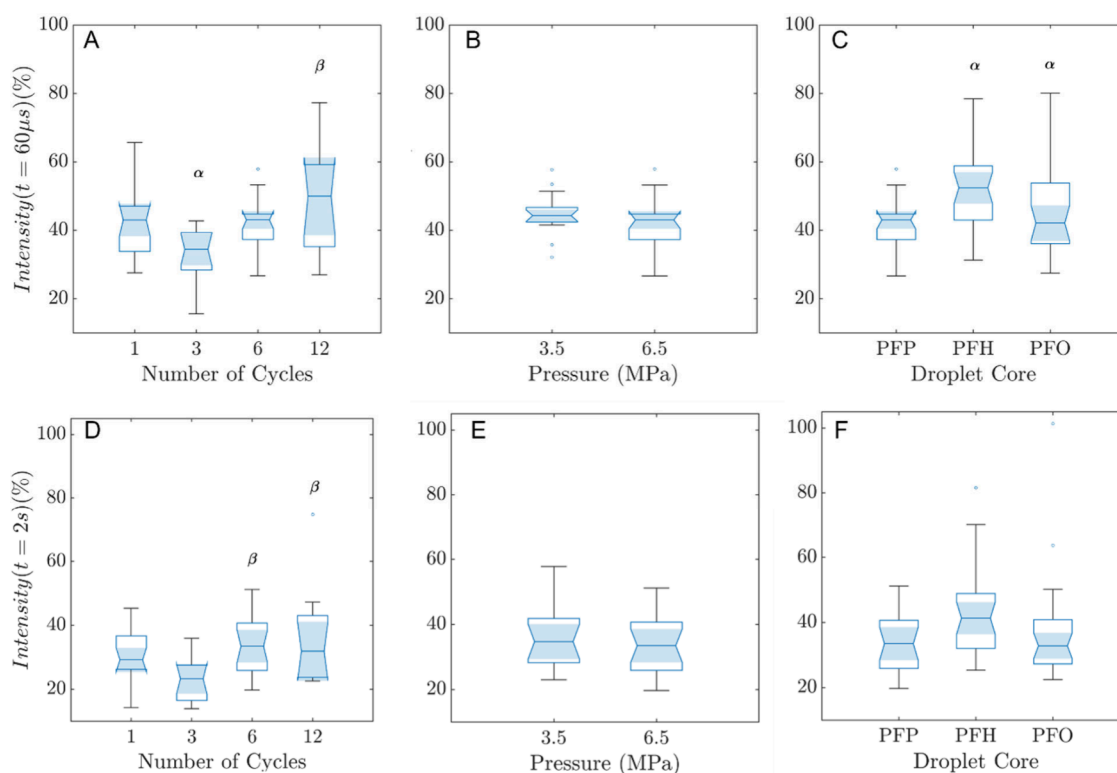


Figure 4. Influence of ultrasound parameters and droplet composition on the normalized payload fluorescence intensity at 60 μ s and 2 s after ADV. Normalized fluorescence intensity (%) measured 60 μ s after ADV as a function of (A) the number of ultrasound cycles at 6.5 MPa in PFP droplets ($N = 14$), (B) peak negative pressures using a 6-cycle burst ($N = 16$), and (C) droplet core composition ($N = 21$) at 6.5 MPa and a 6-cycle burst. (d through F) Corresponding fluorescence intensity values measured 2 s after ADV. Box plots represent the median, interquartile range, and outliers for each condition. Statistically significant differences ($p < 0.05$) are denoted as follows: (A and D) α , vs 1 cycle; β , vs 3 cycles; (C) α , vs PFP.

samples were polymerized directly within a 3D-printed holder and formed into disks approximately 3 mm thick and 14 mm in diameter (Figure 1B). The transducer was optically aligned with a 40 $\times$ water immersion objective. Ultrasound was delivered through a water-filled coupling cone, as shown in Figure 1. ADV was induced using single ultrasound bursts containing 1 to 12 cycles, at a driving frequency of 2.5 MHz. Two peak negative pressures were tested: 3.5 and 6.5 MPa. Although the frequency and driving pressure have been shown to significantly affect the acoustic responses of conventional microbubbles, the pulse duration determines the time period over which the generated bubbles could exert a desired bioeffect. To capture acoustic emissions from individual ADV-generated bubbles, a passive cavitation detection technique was implemented using a broadband hydrophone (HNC1000, Onda, Sunnyvale, CA, USA; Figure 2B). A total of 15 individual time traces were recorded per condition at a sampling rate of 1 Gsamples/s via an oscilloscope (model HDO4034, Teledyne LeCroy). These signals were analyzed offline by using MATLAB to compare both temporal and spectral responses of ADV bubbles. Unless otherwise specified, pressures are reported as peak rarefactional pressures throughout the paper. In our previous work, we showed that standing-wave interference was limited to ~ 1 mm from a free interface using the same transducer.²⁶ The tight focus (f -number: 0.8) and use of single ultrasound bursts minimized standing-wave formation at the gel–coverslip interface. Thus, in this study, data were collected at the midthickness of fibrin-based hydrogels (~ 1 mm above the coverslip).

2.6. Statistical Analysis. Statistical analyses were conducted in MATLAB using one-way analysis of variance (ANOVA) followed by Tukey's multiple comparisons test to assess differences across experimental conditions. Results are presented as box plots, illustrating the median, interquartile range, and outliers for each condition. The number of independent replicates and statistical significance levels are provided in the corresponding figure captions.

Statistical significance was defined as $p < 0.05$, corresponding to a 95% confidence level.

3. RESULTS AND DISCUSSION

3.1. Ultrahigh-Speed Fluorescence Microscopy for Payload Release Dynamics. Real-time dynamics of the ADV-triggered payload release were recorded in fibrin-based hydrogels containing PFP phase-shift droplets. From this, we isolated the evolution of the fluorescence payload over a 2-s period (Figure 3). Droplets initially loaded with a homogeneously distributed fluorescent payload underwent vaporization using a six-cycle burst of 2.5 MHz focused ultrasound at a peak negative pressure of 6.5 MPa, initiating rapid payload release (Figure 3A). Within the microsecond regime, fluorescence imaging revealed a sharp radial expansion of the fluorescence signal, indicative of rapid, advection-driven payload dispersal induced by ADV bubble formation and expansion. Imaging was subsequently continued over millisecond-to-second time scales to track the redistribution of the released payload, allowing assessment of slower, diffusion-dominated transport following ADV (Figure 3B).

The fluorescence intensity, normalized to the initial pre-ADV signal, was quantified across frames (Figure 3C,D). Additionally, the radial transport distance of the payload front (d)—normalized to the initial droplet radius (R_{init})—was calculated for each frame to assess the spatial distribution of the payload in each frame. During the first few microseconds of ultrasound exposure, a steep decline in the normalized fluorescence intensity from 100% to approximately 30% was

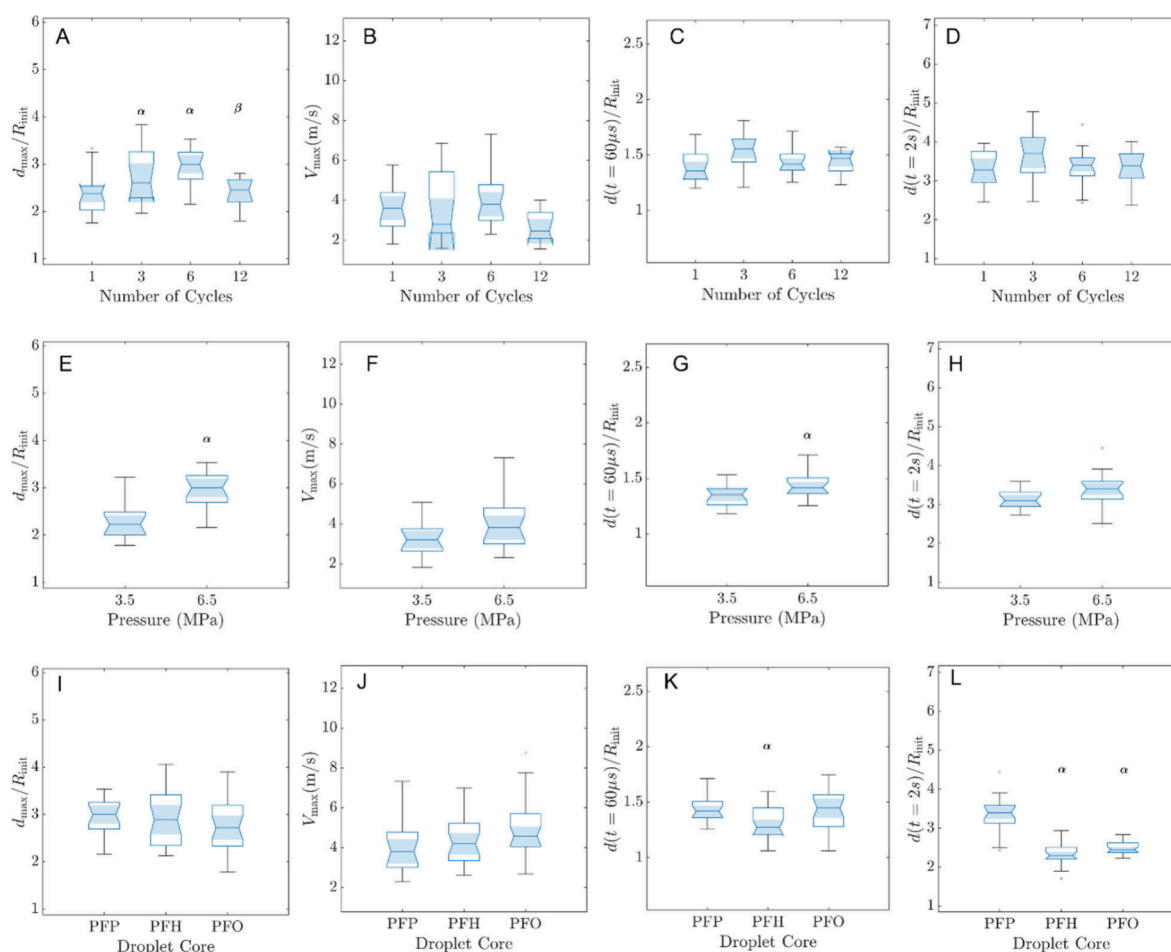


Figure 5. Effect of ultrasound parameters on the payload release dynamics. The maximum release distance normalized by the initial droplet radius ($d_{\max}/R_{\text{init}}$), maximum release velocity ($V_{\max}$), and release distances measured at $60 \mu\text{s}$ [$d(t = 60 \mu\text{s})$] and 2 s [$d(t = 2 \text{ s})$] post-ADV were plotted as a function of (A–D) the number of ultrasound cycles ($N = 14$), (E–H) peak rarefactional pressure ($N = 16$), and (I–L) different PFC liquid cores ($N = 21$). Box plots indicate the median, interquartile range, and outliers for each condition. Statistically significant differences ($p < 0.05$) are denoted as follows: (A–D) α , vs 1 cycle; β , vs 6 cycles; (E–H) α , vs 3.5 MPa; (I–L) α , vs PFP.

observed, corresponding to $\sim 70\%$ payload release (Figure 3C). This phase coincided with a rapid increase in the radial transport distance of the payload. Once ultrasound was turned off, the payload continued to disperse at a markedly slower rate, suggesting a transition from acoustically dominated advection to diffusion-driven transport (Figure 3D). Notably, the ADV bubble persisted after vaporization, as evidenced by the absence of fluorescence within its interior. Over the 2-s observation period, the fluorescence intensity decreased to below 20%, indicating that over 80% of the initial payload had either diffused away or redistributed beyond the imaging field. At time points beyond 1 s, fluorescence signals became nearly undetectable, confirming near-complete payload release. To assess the potential for fluorescence self-quenching, droplets were formulated with varying concentrations of Alexa Fluor 647-labeled dextran in the W_1 phase (0.06–3.3 mg/mL; Figure 3E). The fluorescence intensity increased with an increasing W_1 concentration, with no evidence of signal saturation within the tested range. The concentration used in high-speed imaging studies (1.7 mg/mL) lies below this upper bound.

High-speed fluorescence microscopy showed that even a single cycle of focused ultrasound, at the pressure and frequency studied here, was sufficient to induce substantial (near-complete) payload release in both PFP and PFH

droplets (Figure 4). This response is attributed to the stable and irreversible vaporization behavior of phase-shift droplets under these suprathreshold conditions. For the droplet sizes used, ADV threshold pressures have previously been reported as $\sim 1.5 \text{ MPa}$ (PFP), $\sim 1.8 \text{ MPa}$ (PFH), and $\sim 2.5 \text{ MPa}$ (PFO).²⁵

While increasing the number of cycles did not yield a consistent trend in overall release percentage, distinct differences were observed in the spatial characteristics of release. At lower cycle numbers, payload release appeared more symmetrical and localized near the bubble interface, whereas at 12 cycles, the release pattern was more chaotic and inhomogeneous (see Videos S1 and S2). In particular, increased in-plane release of fluorescent payload—both as diffuse cloud and vesicular fragments—was evident at longer pulse durations, leading to elevated fluorescence intensity measurements at both microsecond and second time scales (Figure 4A,D).

Next, the influence of the driving pressure on the release intensity was assessed in PFP droplets (Figure 4B,E). Both 3.5 and 6.5 MPa exceeded the ADV threshold and resulted in substantial release, with no statistically significant difference in the fluorescence intensity at either the $60 \mu\text{s}$ or 2 s time points. The role of droplet composition was investigated by comparing PFP, PFH, and PFO droplets, all exposed to 6.5 MPa and 6-

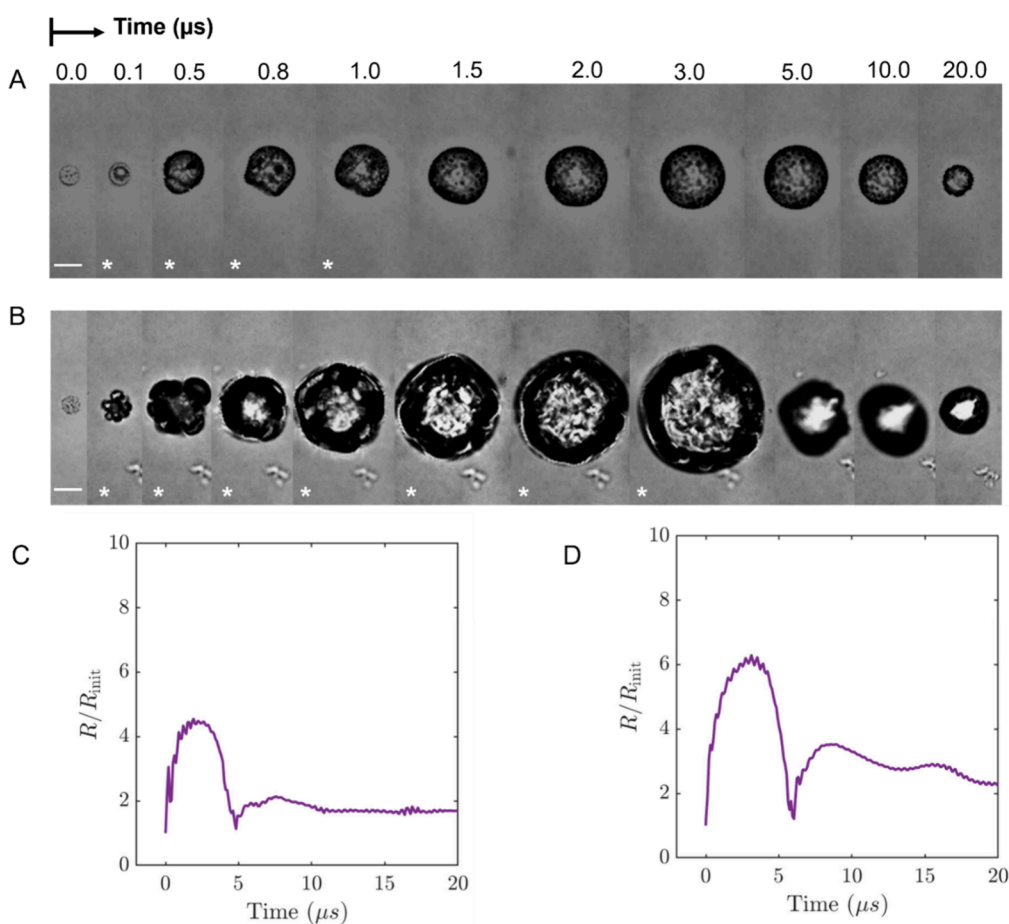


Figure 6. High-speed brightfield microscopy images capturing the temporal evolution of bubbles generated via ADV. Sequential high-speed images showing bubble formation and expansion in response to (A) 3-cycle and (B) 12-cycle ultrasound at 2.5 MHz and a peak negative pressure of 6 MPa. Asterisks (*) indicate the presence of ultrasound. Scale bar: 15 μm . Corresponding normalized bubble radius (R/R_{init}) plotted as a function of time for (C) 3-cycle and (D) 12-cycle ultrasound single bursts.

cycle bursts (Figure 4C,F). At the microsecond time scale, PFP droplets exhibited significantly lower post-ADV fluorescence intensity compared to PFH and PFO, indicating more release during the initial ADV event (Figure 4C). However, at 2 s, these differences diminished, and the fluorescence intensity was comparable across all PFC formulations (Figure 4F).

To assess the influence of ultrasound parameters and droplet composition on payload release dynamics, several quantitative metrics were evaluated: the normalized maximum release distance during ultrasound exposure ($d_{\text{max}}/R_{\text{init}}$), maximum release velocity during ultrasound (V_{max}), and normalized payload transport distances 60 μs [$d(t = 60 \mu\text{s})/R_{\text{init}}$] and 2 s [$d(t = 2 \text{ s})/R_{\text{init}}$] post-ADV (Figure 5). An increase in the ultrasound number of cycles led to a corresponding rise in the normalized maximum release distance (Figure 5A). For example, a 6-cycle pulse resulted in a $\sim 30\%$ increase in $d_{\text{max}}/R_{\text{init}}$ compared to a single cycle. However, higher cycle numbers also produced more chaotic and heterogeneous release patterns. Despite this, the number of cycles did not significantly affect the maximum release velocity during ADV (Figure 5B).

At 60 μs and 2 s after ultrasound, release distances reached up to ~ 1.5 and ~ 3.5 times the initial droplet radius, respectively, indicating that most of the payload dispersed rapidly during the early phase, with continued redistribution occurring over a longer time scale (Figure 5C,D). Increasing

the peak negative pressure from 3.5 to 6.5 MPa produced an $\sim 36\%$ increase in $d_{\text{max}}/R_{\text{init}}$ (Figure 5E), although the release velocity was not significantly affected (Figure 5F). To examine the effect of the droplet composition, we compared PFP, PFH, and PFO droplets, all exposed to identical acoustic conditions (2.5 MHz, 6 cycles, 6.5 MPa). While $d_{\text{max}}/R_{\text{init}}$ values during ultrasound were similar among all droplet types (Figure 5I), the payload release distance at 2 s post-ADV was significantly greater for PFP droplets—approximately 47% higher than that of PFH and 54% higher than that of PFO (Figure 5L). This is likely attributed to the more rapid growth of ADV bubbles in PFP droplets due to their lower bulk boiling point. Saturation vapor pressures of ~ 135 , ~ 47 , and ~ 0.8 kPa have been reported for PFP, PFH, and PFO liquids, respectively.^{45,57} No significant differences in release velocities were observed during ultrasound exposure under any of the conditions studied here. In addition, average velocities measured on millisecond time scales ranged from 2.7 to 4.2 $\mu\text{m/s}$ across all conditions, with no significant dependence on the ultrasound cycle number or pressure. However, when droplet formulations were compared, PFP droplets exhibited significantly higher average release velocity ($2.7 \pm 1.0 \mu\text{m/s}$) compared to PFH ($0.9 \pm 0.5 \mu\text{m/s}$) and PFO ($0.6 \pm 0.4 \mu\text{m/s}$) on the millisecond time scale.

A previous study reported payload release velocities of ~ 0.6 m/s at 5 MHz and 12 MPa from micron-sized PFP droplets

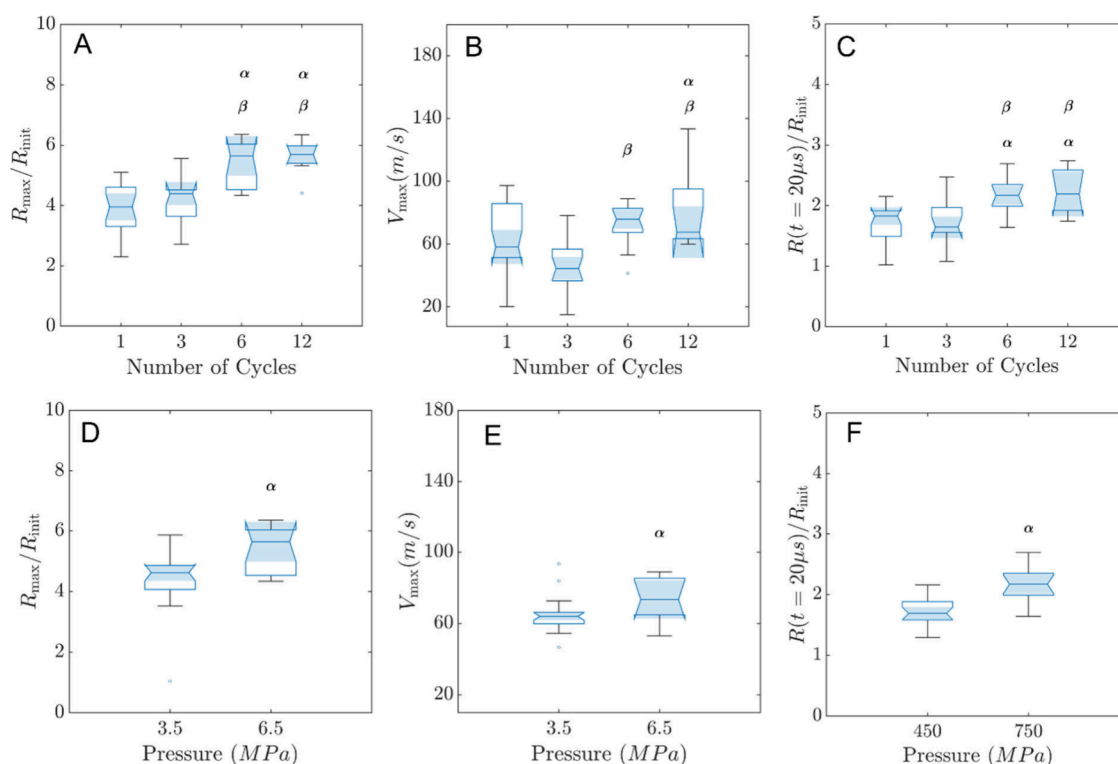


Figure 7. Quantification of ADV bubble dynamics from high-speed microscopy images. The effect of the ultrasound cycle number on (A) maximum bubble expansion ($R_{\max}/R_{\text{init}}$), (B) maximum bubble expansion velocity ($V_{\max}$), and (C) bubble radius at $20\ \mu\text{s}$ post-ADV [$R(t = 20\ \mu\text{s})$] was analyzed ($N = 13$). (D–F) Corresponding measurements as a function of the peak rarefactional pressure ($N = 13$). Statistically significant differences ($p < 0.05$) are denoted as follows: (A–C) α , vs 1 cycle; β , vs 3 cycles; (D–F) α , vs 3.5 MPa.

using lower-frame-rate fluorescence microscopy (100 kfps).²⁸ In that work, vesicle release was observed only at higher driving pressures (12 MPa), and the results were influenced by droplet proximity to solid boundaries, such as cellulose tube walls, which may have constrained dispersal. In contrast, droplets located farther from boundaries exhibited cloud-like fluorescent release patterns, as observed in our experiments. Similar findings were reported in studies using liposome-loaded microbubbles, where vesicle release required long ultrasound pulses (~ 1000 cycles) at low pressures (~ 20 kPa), whereas cloud-like release occurred only at higher pressures (>60 kPa) and shorter durations.⁵⁸

In our study, the maximum payload release velocity during ADV was measured at 2–4 m/s (Figure 5), which exceeds previous reports for both vesicle (~ 0.1 m/s) and cloud (~ 1 m/s) release from conventional microbubbles in Newtonian fluids.^{48,58} For reference, the velocity field generated by acoustic microstreaming around oscillating microbubbles is typically on the order of 0.01–0.1 m/s.⁵⁰ Moreover, while conventional microbubble systems often require long-duration pulses to achieve complete release,^{48,59} near-complete payload release in our study was achieved with significantly shorter bursts (Figure 4).

Since the ultrasound frequency (2.5 MHz) exceeded the frame rate of high-speed fluorescence imaging (2 Mfps), rapid payload motion during ADV may have introduced motion blur, limiting resolution during the initial release. It is important to note that fluorescence intensity measurements may also be influenced by displacement of the released payload outside the imaging plane due to bubble expansion or dilution of the fluorescent dye within the fibrin hydrogel matrix. Additionally, the small size of the released payload from the W_1

phase may have been below the detection limit of the imaging system, appearing as a diffuse fluorescence cloud near the bubble interface. Future studies will explore the use of more photostable and brighter fluorescent markers, such as quantum dots or dye-loaded nanoparticles, to improve the payload detection sensitivity, particularly at high frame rates. Several prior studies have shown that both microbubble cavitation and droplet vaporization can result in directional payload transport, predominantly along the axis of the ultrasound beam. This directionality is largely driven by acoustic radiation forces and streaming. As a result, imaging limited to a single focal plane may underestimate or entirely miss axial payload delivery.^{60–62}

3.2. High-Speed Microscopy for ADV Bubble Dynamics. The dynamic response of phase-shift droplets undergoing ADV was characterized using ultra-high-speed brightfield microscopy at 10 Mfps (Figure 6). ADV initiation detected by the appearance of a high-contrast vapor bubble manifested initially by a change in gray-scale contrast inside the droplet. Representative time-lapse images show the evolution of ADV bubbles triggered by 3-cycle (Figure 6A) and 12-cycle (Figure 6B) ultrasound bursts at 2.5 MHz and a peak negative pressure of 6.5 MPa. The corresponding radial dynamics, where the bubble radius (R) is normalized by the initial droplet radius (R_{init}), are plotted in Figure 6C (3 cycles) and Figure 6D (12 cycles). These curves reveal a rapid initial expansion, followed by oscillatory behavior driven by the compressional and rarefaction phases of the ultrasound burst. The bubbles reached their maximum size, while ultrasound was still being applied. The elasticity of the surrounding fibrin hydrogel stored potential energy during ADV bubble expansion, which was then released after ultrasound was turned off, leading to subsequent contraction of the bubble.⁶³ Accurately capturing

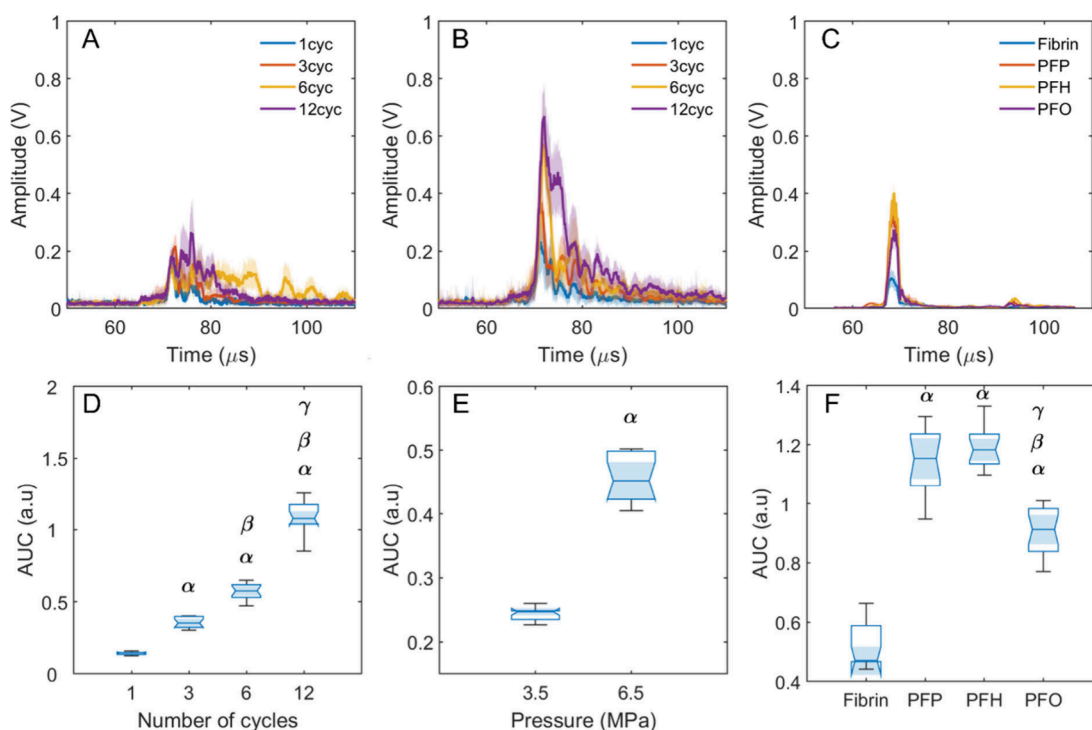


Figure 8. Acoustic emissions from ADV bubbles were recorded by using a passive cavitation detection technique. Individual voltage–time traces of acoustic emissions were recorded and averaged to assess the effect of ultrasound parameters. Amplitude envelope of acoustic signals from (A) fibrin-only hydrogels (control) and (B) fibrin hydrogels containing PFP droplets exposed to an ultrasound burst at 2.5 MHz (6.5 MPa) with varying numbers of cycles. (C) Comparison of droplets with different PFC cores, exposed to ultrasound at 2.5 MHz, 6.5 MPa, and a constant number of ultrasound cycles: 6. AUC of acoustic emissions analyzed as a function of the (D) number of ultrasound cycles, (E) peak negative pressure, and (F) droplet core composition while fibrin-only gels were included as a control. Statistically significant differences ($p < 0.05$) are denoted as follows: (D) α , vs 1 cycle; β , vs 3 cycles; γ , vs 6 cycles; (E) α , vs 3.5 MPa; (F) α , vs fibrin; β , vs PFP; γ , vs PFH. $N = 15$ for all conditions.

the moment and size of the minimum radius remains challenging due to the high temporal resolution required. In our experiments, the resulting bubble radius consistently remained greater than the initial droplet radius ($R/R_{\text{init}} > 1$), likely due to the high vapor pressure of the PFC core, which resists full collapse and dissolution. Radius–time curves of individual droplets in this study consistently exhibited distinct phases (Figure 6): an initial rapid vapor bubble expansion, followed by oscillatory growth during ultrasound exposure, with bubble wall velocities on the order of 50–70 m/s for 12 μm droplets. This was typically followed by a collapse phase and unforced oscillatory decay toward a quasi-equilibrium size once ultrasound was turned off. In contrast, previous studies conducted under elevated temperatures (e.g., 41–50 $^{\circ}\text{C}$) or higher degrees of superheat have reported monotonic, asymptotic bubble growth without collapse—particularly in superheated PFP droplets larger than 6 μm in diameter.^{43,64}

The maximum expansion ratio during ultrasound (i.e., $R_{\text{max}}/R_{\text{init}}$) increased with ultrasound pulse duration. For example, bubbles formed with 6-cycle bursts expanded 44% more than those formed with a single cycle (Figure 7A). This expansion may arise from either progressive vapor growth via rectified diffusion or multiple vaporization events within the droplet. Previous studies have shown that short pulses (e.g., 2 cycles) only partially vaporize micron-sized PFP droplets, whereas longer pulses (e.g., 13 cycles) can phase-convert a larger fraction of the core.⁶⁵ In our present study, further increasing the pulse duration beyond 6 cycles did not significantly enhance expansion, suggesting that bubble growth may

become limited by the elasticity of the surrounding fibrin hydrogel. Consistent with this, the maximum expansion ratio of ADV bubbles exposed to 24-cycle ultrasound was not significantly different from those generated with 6- or 12-cycle bursts (data not shown). Furthermore, radial Hencky strain [$\ln(R/R_{\text{init}})$],⁵⁶ evaluated at maximum bubble expansion, ranged from 1.3 to 1.7. The ADV-generated bubbles reached their peak expansion at 2.6 ± 1.0 , 3.6 ± 1.6 , 2.99 ± 0.9 , and 3.5 ± 0.9 μs for 1-, 3-, 6-, and 12-cycle ultrasound pulses, respectively. These corresponded to ultrasound pulse durations of 0.4, 1.2, 2.4, and 4.8 μs , respectively.

Increasing the pressure from 3.5 to 6.5 MPa led to a 22% increase in the maximum expansion ratio, a 12% increase in the expansion velocity, and a 24% increase in the resulting bubble diameter measured at 20 μs post-ADV (Figure 7D–F). A comparison of the relative expansion [$E = (R_{\text{max}} - R_{\text{init}})/R_{\text{init}}$] and compression [$C = (R_{\text{min}} - R_{\text{init}})/R_{\text{init}}$] across experiments revealed that ADV bubbles exhibited an expansion-dominated oscillation profile, with $E/C \gg 2$, indicative of higher resistance to compression than expansion. For comparison, symmetric oscillations typically yield values of E/C close to 1, while ratios below 0.5 are characteristic of compression-dominated behavior.^{41,66} These findings imply that due to their internal pressure, large size, and inertia of the surrounding hydrogel matrix, ADV-generated bubbles are less responsive to the short compressional phases of the ultrasound pulse.

In our earlier work, we showed that higher-boiling-point phase-shift droplets, such as PFO, underwent transient bubble formation during ultrasound, recondensation, and fragmenta-

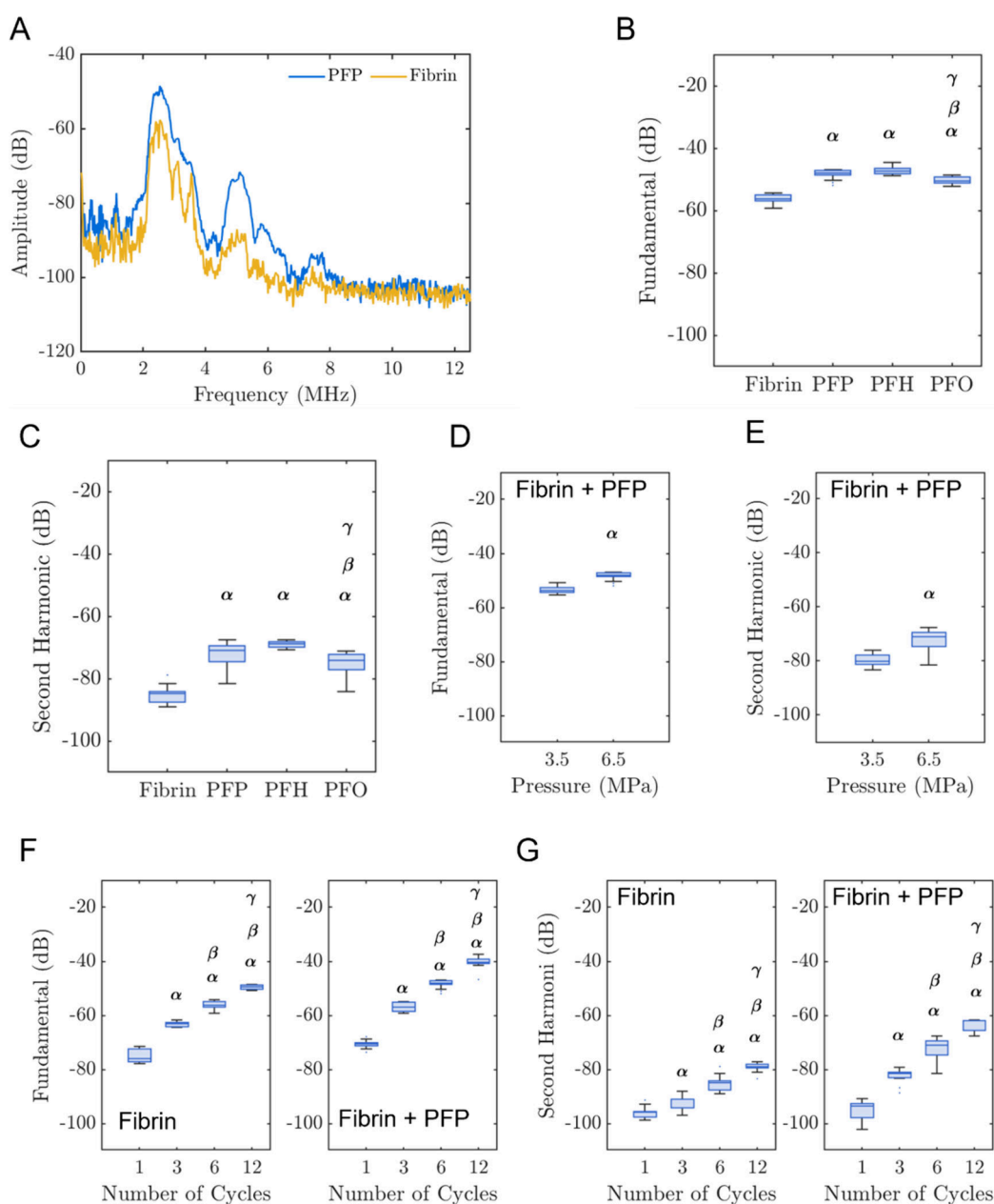


Figure 9. Acoustic emissions from ADV bubbles in fibrin hydrogels under varying ultrasound parameters and droplet core compositions. (A) Representative frequency spectra comparing fibrin-only controls and fibrin hydrogels containing PFP droplets exposed to ultrasound at 2.5 MHz, 6.5 MPa, and 6 cycles. Quantification of (B) fundamental and (C) second-harmonic amplitudes for droplets with different PFC cores (PFP, PFH, and PFO) under identical ultrasound conditions (2.5 MHz, 6.5 MPa, and 6 cycles). Fibrin-only hydrogels were used as controls. Effect of pressure on (D) fundamental and (E) second-harmonic emissions from PFP droplets exposed to 2.5 MHz and 6 ultrasound cycles. Effect of ultrasound number of cycles on the (F) fundamental and (G) second-harmonic emissions in fibrin-only hydrogels and fibrin hydrogels containing PFP droplets (2.5 MHz, 6.5 MPa). Statistically significant differences ($p < 0.05$) are denoted as follows: (B and C) α , vs fibrin; β , vs PFP; γ , vs PFH; (D and E) α , vs 3.5 MPa; (F and G) α , vs 1 cycle; β , vs 3 cycles; γ , vs 6 cycles. $N = 15$ for all conditions.

tion after ultrasound exposure, leaving behind fluid-filled macropores.^{45,67} Although the precise timing of these fragmentation events could not be resolved in our studies, prior investigations have reported that such processes typically occur within hundreds of picoseconds after peak compression in conventional microbubbles.⁶⁸ A comparison of ADV bubble dynamics (Figure 6C,D) with fluorescence-based measurements of payload release (Figure 3C,D) suggests a direct mechanistic link between the bubble behavior and payload dispersal. The ratio of $R_{\max}/R_{\text{init}}$ and $d_{\max}/R_{\text{init}}$ exceeded unity across all experimental conditions. It is important to note that

bubble sizes appear larger in brightfield images than in fluorescence microscopy due to optical diffraction and scattering effects in brightfield imaging, while fluorescence signals originate solely from labeled payload molecules.⁵⁸

To further explore the coupling between bubble dynamics and payload transport, we calculated the Stokes number, a dimensionless parameter representing the ratio of the particle response time to fluid time scale. For ultrasound at 2.5 MHz, the excitation period is approximately 0.4 μs . The estimated response time of submicron payload particles is $\sim 0.03 \mu\text{s}$,⁵⁸ yielding a Stokes number of ~ 0.07 . A Stokes number of less

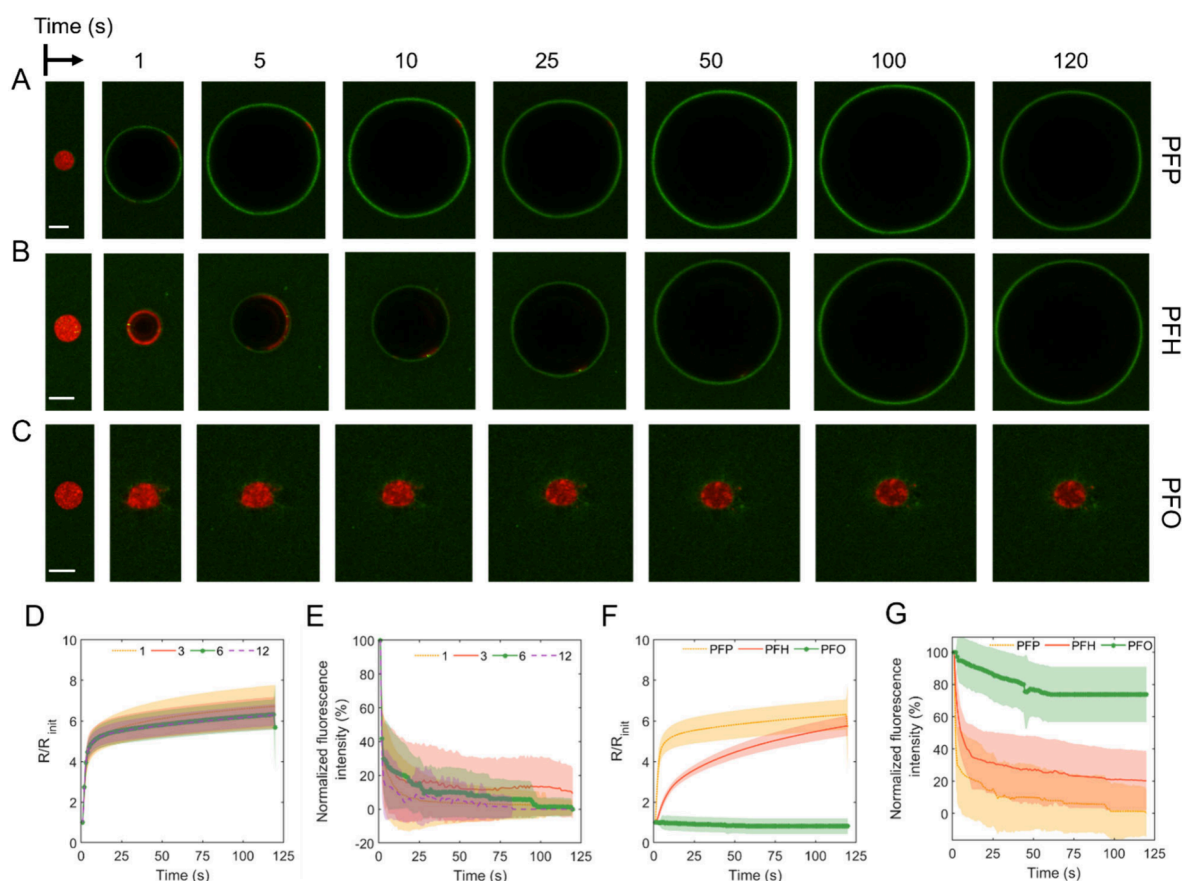


Figure 10. Longitudinal diffusion-driven growth of ADV bubbles and corresponding payload release dynamics in phase-shift droplets. Time-lapse confocal fluorescence microscopy was used to monitor ADV bubble expansion and payload dispersal in fibrin-based tissue-mimicking hydrogels. Representative images are shown for (A) PFP, (B) PFH, and (C) PFO droplets following ultrasound exposure (2.5 MHz, 6.5 MPa, single burst of 6 cycles). Green fluorescence corresponds to the labeled fibrin matrix; red fluorescence indicates an encapsulated payload. Scale bar: 15 μm . (D) Expansion ratio (R/R_{init}) of PFP droplets as a function of the applied number of ultrasound cycles at 2.5 MHz and 6.5 MPa. (E) Corresponding normalized fluorescence intensity profiles over time, representing ADV-induced payload release for the same ultrasound conditions in panel D. (F) Comparison of expansion ratio and (G) normalized fluorescence intensity comparing droplets with different PFC cores (2.5 MHz, 6.5 MPa, single burst of 6 cycles). Data represent an average of 25 droplets for each condition. The shaded region represents the standard deviation.

than 1 indicates that particles closely follow fluid motion. This tight coupling suggests that bubble-induced advection plays a dominant role in the initial payload dispersal, with diffusion taking over at later time points. In the diffusion-dominated regime though, as is shown in section 3.4, the payload release rate is significantly faster than the bubble growth rate.

3.3. Analysis of Acoustic Emissions from Individual Phase-Shift Droplets. In ultrasound imaging and therapeutic applications, the nonlinear oscillations of gas-filled microbubbles are central to their contrast-enhancing capabilities,^{69,70} in situ blood pressure monitoring,^{71–74} and targeted therapeutic applications.^{5,75} To characterize the acoustic response of droplets undergoing ADV, passive cavitation detection was employed in parallel with optical microscopy. Figure 8 displays the average amplitude envelopes of acoustic emissions recorded in fibrin hydrogels containing droplets exposed to ultrasound at 2.5 MHz and a peak negative pressure of 6.5 MPa. Control experiments using fibrin-only hydrogels provided a baseline for comparison.

Despite using a dilute droplet concentration [i.e., 0.01% (v/v)], the hydrophone likely captured the acoustic emissions originating from multiple vaporization events occurring simultaneously within the ultrasound focal volume. To quantify acoustic emissions, the area under the curve (AUC)

of the averaged amplitude envelopes was calculated. As shown in Figure 8D, an increase in the number of ultrasound cycles led to enhanced acoustic emissions, likely due to cumulative rectified diffusion effects that resulted in larger oscillation amplitudes and, thus, greater acoustic emissions. Similarly, increasing the peak negative pressure from 3.5 to 6.5 MPa significantly elevated acoustic output (Figure 8E). When droplet formulations were compared, PFP and PFH droplets generated stronger acoustic emissions (Figure 8F).

The frequency spectra of the acoustic emissions, computed using fast Fourier transform, revealed strong fundamental and second-harmonic components in droplets undergoing ADV, with signal amplitudes substantially higher than those recorded in fibrin-only controls (Figure 9A). Both fundamental and second-harmonic amplitudes were slightly but significantly lower for PFO droplets compared to PFP and PFH under similar acoustic conditions (Figure 9B,C). With increased driving pressure and pulse duration, the generated ADV bubbles can oscillate asymmetrically, resulting in the scattering of energy to harmonic multiples of the transit frequency. These nonlinear responses were strongly dependent on both pulse duration and peak negative pressure: longer ultrasound bursts and higher pressures significantly increased both fundamental and second-harmonic amplitudes in the PFP droplets (Figure

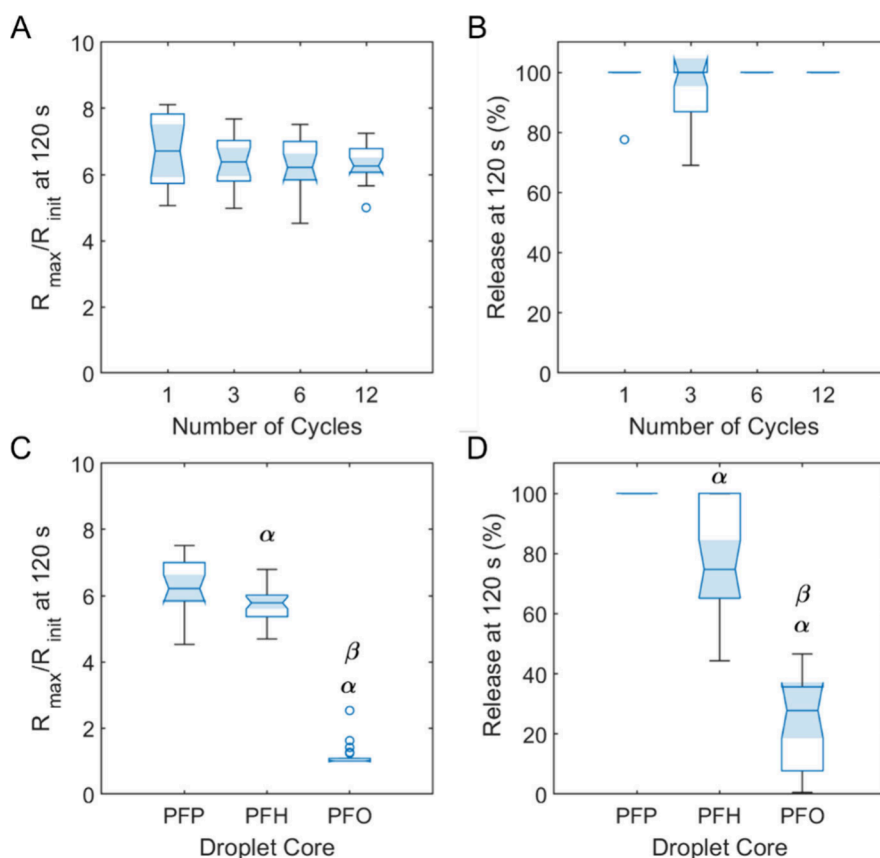


Figure 11. Long-term expansion and payload release of ADV-generated bubbles at 120 s postultrasound recorded via confocal imaging. (A) Maximum normalized expansion ratio and (B) total percentage of payload released at 120 s as a function of the number of cycles. (C and D) Comparison of these parameters for droplets with different PFC cores. Statistically significant differences ($p < 0.05$) are denoted as follows: α , vs PFP; β , vs PFH. $N = 25$ for all conditions.

9D–G). However, under single-cycle excitation, second-harmonic amplitudes were not significantly different from those observed in fibrin-only controls. Interestingly, subharmonic emissions (at half the driving frequency) did not show significant variations across the tested conditions (data not shown). This may be attributed to saturation effects at higher pressures⁷¹ or interference patterns arising from reflections within the experimental setup. Unlike higher harmonics, which typically scale with the square of the incident pressure, subharmonic emissions are threshold-dependent and highly sensitive to the size distribution of the generated bubbles.⁷³

For optimal imaging in optical microscopy, larger micrometer-sized droplets were used in this study, which would generate bubbles of tens (during) or hundreds (post) of micrometers due to ADV. However, such large bubbles are generally poor nonlinear scatterers, which may have influenced the harmonic content observed in this study. Despite efforts to minimize acoustic reflections, such as angling the transducer and hydrophone relative to the sample, interference patterns were still possible due to reflections at the gel–coverslip interface. These artifacts may resemble in vivo boundary effects, where ADV bubbles encounter tissue interfaces, bone, or implanted materials.

3.4. Time-Lapse Confocal Fluorescence Imaging.

Time-lapse confocal fluorescence microscopy was utilized to study the long-term evolution of ADV-generated bubbles and the corresponding release of encapsulated payloads up to 120

s. This prolonged observation window was crucial for capturing slower processes, such as gas-diffusion-driven bubble expansion and payload transport, which are particularly relevant for sustained drug delivery and tissue engineering applications. During the post-ADV phase, bubble growth was largely governed by gas diffusion from the surrounding hydrogel, continuing until equilibrium was reached between the internal bubble gas pressure and the surrounding medium. Unlike conventional microbubbles, which typically fragment or dissolve at pressures above 200 kPa,^{39,76} the ADV-generated bubbles in this study remained stable even under megapascal-range ultrasound exposure (Figure 10).

Bubble expansion was quantified by normalizing the radius of the bubble over time to the initial droplet radius, while payload release was assessed by tracking the fluorescence intensity normalized to pre-ADV levels (Figure 10). The normalized bubble radius rapidly increased during the initial period post ultrasound and gradually stabilized as it approached an equilibrium driven by gas diffusion and hydrogel elasticity. The fluorescence intensity sharply declined within the first few seconds, indicating substantial payload release initiated during ADV and subsequently sustained by diffusion-based transport mechanisms. Notably, the number of ultrasound cycles had minimal influence on the final bubble expansion or final fluorescence intensity at 120 s (Figure 11A,B). These metrics represent the end point of time-lapse confocal observations and reflect the cumulative effects of both acoustic and diffusion-driven processes. ADV bubbles formed

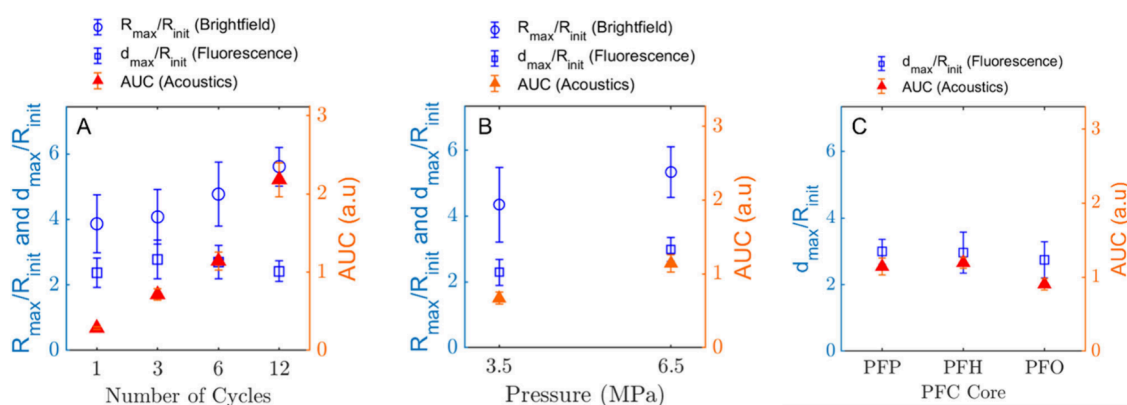


Figure 12. Comparison of ADV dynamics, release dynamics, and acoustic emissions as a function of the (A) ultrasound number of cycles, (B) pressure, and (C) PFC liquid core.

from PFP droplets—which have the lowest boiling point among tested PFCs—exhibited significantly greater expansion compared to those formed from PFH or PFO droplets (Figure 11C). PFO droplets, with the highest bulk boiling point, displayed a partial payload release, followed by recondensation (Figure 11D).

Figure 12 provides a comparative analysis of metrics derived from different measurement modalities, high-speed fluorescence imaging ($d_{\max}/R_{\text{init}}$), high-speed brightfield imaging ($R_{\max}/R_{\text{init}}$), and acoustic emissions (AUC). These metrics are plotted as a function of the number of cycles, peak negative pressure, and PFC core composition. Pearson correlation analysis revealed a strong positive correlation ($r = 0.9$) between acoustic emissions and bubble expansion measured via brightfield imaging as a function of the ultrasound cycle number (Figure 12A). This relationship may be explained by the fact that the acoustic scattering intensity in the long-wavelength limit scales with the sixth power of the particle radius.⁷⁷ As shown in Figure 12B, all three metrics increased with a higher pressure, although only two pressure levels were examined. Payload dispersal ($d_{\max}/R_{\text{init}}$) exhibited a strong correlation with acoustic emissions when grouped by PFC composition ($r = 0.8$).

4. CONCLUSIONS

The integration of ultrahigh-speed brightfield and fluorescence microscopy enabled characterization of ADV dynamics and associated payload release across multiple temporal regimes. Our findings indicated that while ultrasound pulse duration and pressure influenced early bubble expansion and emission characteristics, the long-term behavior of ADV bubbles and payload dispersal were predominantly governed by the thermophysical properties of the droplet core. Notably, ADV-generated bubbles produced strong fundamental and second-harmonic emissions, even in response to a single ultrasound burst, underscoring their potential as efficient theranostic agents in ultrasound-mediated drug delivery. Note that these findings are specific to ADV behavior in fibrin-based hydrogels, which serve as a simplified tissue-mimicking environment. While this model allows for controlled observation of bubble dynamics and payload dispersal, it does not capture the full complexity of in vivo drug delivery in the bloodstream. In physiological conditions, additional factors such as convective flow, shear stress, vascular permeability, and clearance mechanisms are likely to play significant roles in determining delivery efficiency and spatial distribution.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.5c11300>.

Ultrahigh-speed fluorescence microscopy visualizing payload release triggered by ADV using single ultrasound bursts containing 1 cycle at a driving frequency of 2.5 MHz (Video S1) (AVI)

Ultrahigh-speed fluorescence microscopy visualizing payload release triggered by ADV using single ultrasound bursts containing 12 cycles at a driving frequency of 2.5 MHz (Video S2) (AVI)

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Notes

The authors declare no competing financial interest.

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